Step 0: Mounting Google Drive and creating the folders

```
from google.colab import drive
drive.mount('/content/drive')

import os
PROJECT_DIR = '/content/drive/MyDrive/doctalk'
os.makedirs(PROJECT_DIR, exist_ok=True)
os.makedirs(f'{PROJECT_DIR}/faiss_index', exist_ok=True)
os.makedirs(f'{PROJECT_DIR}/uploaded_docs', exist_ok=True)

print('Google Drive mounted and project folders ready.')
print(f'Project folder: {PROJECT_DIR}')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
Google Drive mounted and project folders ready.
Project folder: /content/drive/MyDrive/doctalk
```

Step 1: Install Dependencies

```
!pip install -q langchain langchain-community langchain-huggingface
!pip install -q langchain-text-splitters
!pip install -q faiss-cpu
!pip install -q pymupdf
!pip install -q sentence-transformers

print('All dependencies installed.')
```

```
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ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the
google-colab 1.0.0 requires requests==2.32.4, but you have requests 2.33.1 which is incompatible.
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 23.8/23.8 MB 58.8 MB/s eta 0:00:00
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All dependencies installed.
```

Step 2: Uploading Documents

```
from google.colab import files
import shutil

print('Please upload a PDF file...')
uploaded = files.upload()

for filename in uploaded.keys():
    dest = f'{PROJECT_DIR}/uploaded_docs/{filename}'
    shutil.copy(filename, dest)
    PDF_PATH = dest
    print(f'Saved to Drive: {dest}')

print(f'\n Working with: {PDF_PATH}')
```

```
Please upload a PDF file...
Choose Files mars_lithograph  pdf.pdf
mars_lithograph pdf.pdf(application/pdf) - 335116 bytes, last modified: 4/13/2026 - 100% done
Saving mars_lithograph  pdf.pdf to mars_lithograph  pdf.pdf
Saved to Drive: /content/drive/MyDrive/doctalk/uploaded_docs/mars_lithograph  pdf.pdf

 Working with: /content/drive/MyDrive/doctalk/uploaded_docs/mars_lithograph  pdf.pdf
```

Step 3: Loading and Extracting Text

```
from langchain_community.document_loaders import PyMuPDFLoader

loader = PyMuPDFLoader(PDF_PATH)
pages = loader.load()

print(f'Loaded {len(pages)} pages from the PDF')
```

```
print(f'\n--- Preview of Page 1 (first 500 chars) ---')
print(pages[0].page_content[:500])
print(f'\n--- Metadata ---')
print(pages[0].metadata)
```

```
Loaded 2 pages from the PDF

--- Preview of Page 1 (first 500 chars) ---
National Aeronautics and Space Administration
National Aeronautics and Space Administration
300,000,000
900,000,000
1,500,000,000
2,100,000,000
2,700,000,000
3,300,000,000
3,900,000,000
4,500,000,000
5,100,000,000
5,700,000,000
kilometers
0
Mars
www.nasa.gov

--- Metadata ---
{'producer': 'PDFium', 'creator': 'PDFium', 'creationdate': 'D:20260413181446', 'source': '/content/drive/MyDrive/doctalk/upload
```

Step 4: Splitting the text into chunks

```
from langchain_text_splitters import RecursiveCharacterTextSplitter

splitter = RecursiveCharacterTextSplitter(
    chunk_size=500,
    chunk_overlap=50,
    length_function=len
)

chunks = splitter.split_documents(pages)

print(f'Split into {len(chunks)} chunks')
print(f'\n--- Example Chunk ---')
print(f'Content: {chunks[0].page_content}')
print(f'Metadata: {chunks[0].metadata}')
print(f'\n--- Next Chunk (notice the overlap) ---')
print(chunks[1].page_content[:200])
```

```
Split into 17 chunks

--- Example Chunk ---
Content: National Aeronautics and Space Administration
National Aeronautics and Space Administration
300,000,000
900,000,000
1,500,000,000
2,100,000,000
2,700,000,000
3,300,000,000
3,900,000,000
4,500,000,000
5,100,000,000
5,700,000,000
kilometers
0
Mars
www.nasa.gov
Metadata: {'producer': 'PDFium', 'creator': 'PDFium', 'creationdate': 'D:20260413181446', 'source': '/content/drive/MyDrive/doct

--- Next Chunk (notice the overlap) ---
Though details of Mars’ surface are difficult to see from Earth,
telescope observations show seasonally changing features and
white patches at the poles. For decades, people speculated that
bright
```

Step 5: Generate Embeddings with HuggingFace

```
from langchain_huggingface import HuggingFaceEmbeddings

print('Loading embedding model.')

embeddings = HuggingFaceEmbeddings(
    model_name='all-MiniLM-L6-v2',
    model_kwargs={'device': 'cpu'},
    encode_kwargs={'normalize_embeddings': True}
)

# Quick test — embed a single sentence
test_vector = embeddings.embed_query('When did Viking 1 land on Mars?')
print(f'\nEmbedding model loaded!')
print(f'Vector size: {len(test_vector)} dimensions')
print(f'First 5 values: {test_vector[:5]}')
```

```
Loading embedding model.
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
modules.json: 100%  349/349 [00:00<00:00, 15.4kB/s]

config_sentence_transformers.json: 100%  116/116 [00:00<00:00, 6.22kB/s]

README.md:  10.5k/? [00:00<00:00, 551kB/s]

sentence_bert_config.json: 100%  53.0/53.0 [00:00<00:00, 2.33kB/s]

config.json: 100%  612/612 [00:00<00:00, 24.3kB/s]

model.safetensors: 100%  90.9M/90.9M [00:01<00:00, 454MB/s]

Loading weights: 100%  103/103 [00:00<00:00, 364.17it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
tokenizer_config.json: 100%  350/350 [00:00<00:00, 16.6kB/s]

vocab.txt:  232k/? [00:00<00:00, 7.76MB/s]

tokenizer.json:  466k/? [00:00<00:00, 15.6MB/s]

special_tokens_map.json: 100%  112/112 [00:00<00:00, 4.04kB/s]

config.json: 100%  190/190 [00:00<00:00, 9.69kB/s]

Embedding model loaded!
Vector size: 384 dimensions
First 5 values: [0.039327625185251236, 0.024362651631236076, -0.004631285555660725, 0.012242221273481846, 0.0697198212146759]
```

Step 6: Store the embeddings in the vector store

```
from langchain_community.vectorstores import FAISS
from tqdm import tqdm

print(f'Embedding {len(chunks)} chunks and building FAISS index...\n')

# Embed chunks in batches with a progress bar
batch_size = 50
all_embeddings = []

for i in tqdm(range(0, len(chunks), batch_size), desc="Embedding chunks", unit="batch"):
    batch = chunks[i:i + batch_size]
    batch_texts = [chunk.page_content for chunk in batch]
    batch_embeddings = embeddings.embed_documents(batch_texts)
    all_embeddings.extend(batch_embeddings)

print(f'\nEmbedded {len(all_embeddings)} chunks!')
```

```
# Build FAISS index from embeddings
print('Building FAISS index...')
texts = [chunk.page_content for chunk in chunks]
metadatas = [chunk.metadata for chunk in chunks]
vectorstore = FAISS.from_embeddings(
    list(zip(texts, all_embeddings)),
    embeddings,
    metadatas=metadatas
)

# Save to Drive
FAISS_PATH = f'{PROJECT_DIR}/faiss_index'
vectorstore.save_local(FAISS_PATH)

print(f'FAISS index saved to Drive!')
print(f'Location: {FAISS_PATH}')
```

```
Embedding 17 chunks and building FAISS index...

Embedding chunks: 100%|██████████| 1/1 [00:01<00:00,  1.59s/batch]

Embedded 17 chunks!
Building FAISS index...
FAISS index saved to Drive!
Location: /content/drive/MyDrive/doctalk/faiss_index
```

Step 7: Query Test

```
# Modify the query to something relevant to your document
query = "When did Viking 1 land on Mars?"

results = vectorstore.similarity_search(query, k=5)

print(f'Query: "{query}"')
print(f'\nTop {len(results)} most relevant chunks:\n')

for i, doc in enumerate(results):
    print(f'--- Chunk {i+1} (Page {doc.metadata.get("page", "?") + 1}) ---')
    print(doc.page_content)
    print()
```

```
Rings
0


*As of July 2013.
SIGNIFICANT DATES
1877 – Asaph Hall discovers the two moons of Mars.
1965 – NASA’s Mariner 4 sends back 22 photos of Mars, the
world’s first close-up photos of a planet beyond Earth.
1976 – Viking 1 and 2 land on the surface of Mars.
1997 – Mars Pathfinder lands and dispatches Sojourner, the first
wheeled rover to explore the surface of another planet.

--- Chunk 2 (Page 2) ---
Though details of Mars’ surface are difficult to see from Earth,
telescope observations show seasonally changing features and
white patches at the poles. For decades, people speculated that
bright and dark areas on Mars were patches of vegetation, Mars
was a likely place for advanced life forms, and water might exist
in the polar caps. When the Mariner 4 spacecraft flew by Mars in
1965, photographs of a bleak, cratered surface shocked many

--- Chunk 3 (Page 2) ---
Lander was the first mission to “touch” water ice in the martian
arctic. Phoenix also observed precipitation (snow falling from
clouds), as confirmed by Mars Reconnaissance Orbiter. Soil
chemistry experiments led scientists to believe that the Phoenix
landing site had a wetter and warmer climate in the recent past
(the last few million years). NASA’s Mars Science Laboratory
mission, with its large rover Curiosity, is examining martian rocks
```

```
as well. If water ice permeated the entire planet, Mars could

--- Chunk 5 (Page 1) ---
National Aeronautics and Space Administration
National Aeronautics and Space Administration
300,000,000
900,000,000
1,500,000,000
2,100,000,000
2,700,000,000
3,300,000,000
3,900,000,000
4,500,000,000
5,100,000,000
5,700,000,000
kilometers
0
Mars
www.nasa.gov
```